

SONGPAN, China

WMO: 561820

Lat: 32.6709N

Lon: 103.6030E

Elev: 9457

StdP: 10.32

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.0	11.8	-9.0	4.8	30.2	-5.5	5.8	26.1	14.7	38.4	12.4	34.9	3.4	0	0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	77.2	56.9	74.6	56.0	71.9	55.1	59.5	71.7	58.4	69.6	57.3	67.6	6.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.0	95.6	62.2	54.9	91.9	61.7	53.8	88.2	61.5	31.1	72.0	30.2	69.7	29.3	67.7	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.6	11.5	9.7	2.6	82.6	2.8	2.3	0.6	84.3	-1.0	85.6	-2.5	87.0	-4.6	88.7		
(5)			0.9	62.3	2.6	2.0	-0.9	63.7	-2.4	64.8	-3.8	65.9	-5.7	67.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.7	28.3	33.3	39.6	46.2	51.9	56.9	60.9	60.5	55.4	46.7	37.5	29.9		
	DBStd	12.16	4.70	5.53	4.94	4.70	4.37	4.40	3.76	4.00	4.80	4.78	4.81	4.20		
	HDD50	2753	671	468	324	129	30	3	0	0	8	123	374	623		
	HDD65	7075	1136	888	788	565	405	244	135	147	289	566	824	1088		
	CDD50	1171	0	0	1	15	90	210	337	325	171	22	0	0		
	CDD65	14	0	0	0	0	0	1	6	6	1	0	0	0		
	CDH74	280	0	0	0	2	3	26	113	114	22	0	0	0		
	CDH80	13	0	0	0	0	0	1	7	5	0	0	0	0		
	WSAvg	3.4	3.3	3.6	4.0	3.9	3.9	3.2	3.3	3.3	3.2	3.0	2.8	2.9		
	PrecAvg	28.3	0.2	0.4	1.3	2.3	4.1	4.5	4.2	3.5	4.4	2.7	0.5	0.2		
(15) Precipitation	PrecMax	37.0	0.6	1.0	2.4	4.0	6.4	7.3	8.3	6.3	6.6	5.6	1.2	0.8		
	PrecMin	20.7	0.0	0.0	0.2	1.2	1.8	2.0	1.6	1.7	1.7	1.1	0.0	0.0		
	PrecStd	3.2	0.2	0.2	0.6	0.7	1.3	1.3	1.4	1.0	1.4	0.9	0.3	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.7	60.6	64.9	69.9	73.7	77.4	80.6	80.5	76.9	68.3	59.9	52.6		
		MCWB	33.9	37.6	41.1	45.8	51.3	56.2	57.8	58.7	57.0	51.2	40.3	34.6		
	2%	DB	48.7	54.7	59.6	65.6	69.5	72.9	77.1	77.2	72.6	63.3	56.1	48.4		
		MCWB	31.8	35.4	39.5	44.7	50.2	54.8	57.3	56.9	55.1	47.5	38.9	32.3		
	5%	DB	44.7	50.7	55.7	61.9	66.1	69.9	74.3	74.4	68.8	59.7	52.8	45.4		
		MCWB	30.1	34.0	38.4	43.7	48.9	54.0	56.6	56.3	53.5	46.4	37.5	30.9		
	10%	DB	40.6	46.4	51.8	58.3	62.7	66.8	71.1	71.2	65.3	56.4	49.3	41.9		
		MCWB	28.3	32.3	36.9	42.5	48.0	53.3	56.3	55.4	52.5	45.2	36.4	29.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.8	38.6	43.2	49.6	55.5	60.0	60.9	60.8	59.1	53.0	42.8	36.0		
		MCDB	51.0	55.8	60.7	64.1	68.4	71.9	74.0	75.8	72.6	65.3	56.1	50.3		
	2%	WB	32.3	36.4	41.1	47.0	52.7	57.9	59.6	59.2	57.0	50.4	40.4	33.3		
		MCDB	46.9	52.5	55.4	60.2	63.9	67.9	71.6	72.0	68.3	59.9	52.4	46.3		
	5%	WB	30.5	34.6	39.4	45.4	51.0	56.3	58.5	58.0	55.4	48.4	38.9	31.5		
		MCDB	43.3	48.9	52.6	58.0	61.5	65.6	69.4	69.6	65.3	56.7	49.9	43.7		
	10%	WB	28.7	32.8	37.7	43.8	49.3	54.9	57.6	56.9	53.9	46.3	37.3	29.9		
		MCDB	39.9	45.1	49.5	55.4	59.3	63.3	67.3	67.5	62.9	54.1	47.1	41.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	27.9	27.0	24.0	22.8	20.9	18.9	20.4	21.0	19.2	19.0	24.2	27.0		
		MCDBR	34.7	34.0	31.5	30.4	27.6	25.1	26.9	27.3	25.7	24.4	30.0	32.2		
		MCWBR	20.0	17.7	14.3	12.8	11.0	9.4	10.0	9.8	10.3	11.4	15.4	18.2		
		MCDBR	31.8	30.9	27.1	24.9	22.3	19.2	20.2	22.0	21.2	19.8	25.3	29.1		
	5% WB	MCWBR	18.7	16.5	12.7	11.3	9.9	7.7	7.8	8.3	8.7	9.6	13.4	16.9		
(40) Clear-Sky Solar Irradiance	taub		0.305	0.348	0.425	0.443	0.432	0.415	0.430	0.444	0.401	0.346	0.303	0.278		
	taud		2.289	2.153	1.934	1.948	2.070	2.197	2.182	2.109	2.222	2.334	2.378	2.413		
	Ebn at Noon		288	284	268	269	273	276	271	266	273	282	288	294		
	Edh at Noon		33	43	57	59	53	46	47	49	43	35	30	28		
(44) All-Sky Solar Radiation	RadAvg		1092	1266	1485	1635	1607	1443	1567	1478	1207	1082	1071	1026		
	RadStd		72	108	84	114	102	157	184	175	108	108	75	58		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	-2.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+0.54	+0.52	-1.28	N/A	N/A	N/A	-61	-162	+135	+32

Nomenclature: See separate page